

Chapter 11

Public Goods and Externalities

"The curious task of economics is to demonstrate to men how little they really know about what they imagine they can design."

— F.A. Hayek, *The Fatal Conceit*, 1988

Before You Begin: Quick Self-Assessment

Test yourself on these prerequisites before reading. Each answer is in italics.

1. **What is a public good? Give one example.** *A good that is both non-excludable (no one can be prevented from using it) and non-rival (one person's use does not reduce availability to others). Example: national defence, broadcast radio.*
2. **What is the free-rider problem?** *Because a public good cannot exclude non-payers, individuals have an incentive to let others pay and enjoy the benefit for free. The result is systematic private under-provision even when everyone would benefit.*
3. **What is a negative externality? Give one real-world example.** *A cost imposed on a third party outside the market transaction, not reflected in the market price. Example: factory pollution harming downstream residents. The market overproduces because producers do not pay the full social cost.*
4. **What does a Pigouvian tax do?** *It adds a per-unit tax equal to the marginal external cost at the social optimum, forcing producers to internalise the externality. This shifts the effective supply curve from PMC to SMC and moves the market equilibrium to the social optimum.*
5. **What is the key difference between a Pigouvian tax and a cap-and-trade system?** *A Pigouvian tax sets the price of emissions and lets the quantity adjust; cap-and-trade sets the quantity of permits and lets the price adjust. The tax gives price certainty; the cap gives quantity certainty.*

Connecting Week 10 to Week 11: When Markets Fail

Week 10 Established

How strategic firms compete on quantities, prices, and locations. Oligopoly models (Cournot, Stackelberg, Bertrand, Hotelling) all assumed that market participants internalise their own costs and benefits through the price system.

Left open: What happens when prices fail to capture all costs and benefits?

Week 11 Answers It

Public goods: benefits extend to everyone, costs fall on whoever pays. Free-rider problem $\Rightarrow$ private under-provision.

Externalities: costs or benefits spill onto third parties outside the transaction. Overproduction (negative) or underproduction (positive) $\Rightarrow$ dead-weight loss.

Both require policy interventions: property rights, Pigouvian taxes, or cap-and-trade.

The unifying theme: when private prices diverge from social costs and benefits, markets fail. The analysis in this chapter identifies why they fail and which instruments can restore efficiency.

Questions This Chapter Will Answer

- Why does national defence exist only because the government provides it, even though everyone wants it?
- Two neighbours both want a shared streetlight costing \$120. Neither will pay for it alone. How do partial contributions help?
- A factory pollutes a river. Is it better to tax the factory, cap its emissions, or assign property rights?
- Why does the Coase theorem imply that who gets the property right doesn't affect efficiency — only distribution?
- When is a carbon tax better than cap-and-trade, and when is it worse?
- Why do social media platforms give their service away for free, even though the service is enormously valuable?

Learning Objectives and Topics

Each objective states what you will be able to *do*.

LO1 Classify goods using the two-by-two rivalry/excludability taxonomy; explain the free-rider problem.

§11.2 Public Goods

LO2 Analyse private provision of a public good as a game; show how partial contributions transform the prisoner's dilemma into a coordination game.

§11.3 Game-Theoretic Analysis

LO3 Distinguish negative and positive externalities; compute the DWL from overproduction or

underproduction.

§11.4–11.5 Externalities

LO4 Apply the Coase theorem; state its conditions and identify when it breaks down.

§11.6 Solutions

LO5 Compute the Pigouvian tax $t^* = \text{MEC}(Q^*)$ and verify it eliminates DWL.

§11.6 Solutions

LO6 Derive the efficient abatement allocation under cap-and-trade; compare costs to equal regulation.

§11.6 Solutions

Notation Used in This Chapter.

PMC	private marginal cost (supply curve)	MEC	marginal external cost
SMC	social marginal cost = $PMC + MEC$	MB	marginal benefit (demand)
MEB	marginal external benefit	SMB	social marginal benefit = $MB + MEB$
Q^E	competitive equilibrium quantity	Q^*	social optimum quantity
t^*	optimal Pigouvian tax = $MEC(Q^*)$	MAC_i	firm i 's marginal abatement cost
r_i	individual i 's reservation price	g_i	individual i 's contribution

11.1. Introduction

Why does the private market provide too little national defence, too much factory pollution, and too few vaccines — even when people agree these outcomes are wrong?

Two Types of Market Failure — and Three Remedies

Failure	Cause	Direction	Instrument	Key concept
Public goods	Non-excludability	Under-provision	Taxes / mandates	Free-rider problem
Neg. externality	Unpriced cost	Overproduction	Pigou tax / C&T	$SMC > PMC$
Pos. externality	Unpriced benefit	Underproduction	Pigou subsidy	$SMB > MB$

Market failures from public goods and externalities are the most important reasons governments intervene in economies. Climate change — a global public bad with massive negative externalities — illustrates why getting the economics right matters enormously. Every future economist, policy analyst, or business leader will encounter these concepts when grappling with environmental regulation, public infrastructure, healthcare, and innovation policy.

We begin with public goods, then turn to externalities and the policy instruments that correct them.

11.2. Public Goods

Once a lighthouse is built, it guides every ship in the area simultaneously, and one ship's use does not dim the light for others. Why does this seemingly benign property make the lighthouse extremely hard to provide through a market?

11.2.1. Definition: Two Key Properties

Pure Public Good: A good that simultaneously satisfies:

- (i) **Non-excludability:** Once provided, no individual can be prevented from consuming it.

- (ii) **Non-rivalry:** One person's consumption does not reduce availability to others. Marginal cost of serving an additional consumer = 0.

In words: Non-excludability creates the free-rider problem (you can't charge people who will consume it anyway). Non-rivalry means there's no efficiency reason to charge them at all. Together, they make private provision systematically inadequate.

*Non-rival $\neq$ free.
National defence is non-rival but costs hundreds of billions to provide.*

11.2.2. The Goods Taxonomy

The Four-Quadrant Goods Taxonomy

	Rival	Non-rival
Excludable	Private goods Food, clothing, hotel rooms, seats on a plane <i>Market works well: seller can price and exclude</i>	Club goods Netflix, toll roads, gym, national parks with entry fees <i>Excludability $\Rightarrow$ market can price efficiently; MC = 0 for extra user once good exists</i>
Non-excludable	Common-pool resources Fisheries, groundwater aquifers, congested roads <i>Tragedy of the commons: over-exploitation</i>	Pure public goods National defence, broadcast radio, open-source software, public health <i>Free-rider problem $\Rightarrow$ market under-provides</i>

Non-excludability causes the market to fail. Non-rivalry removes the efficiency reason to charge.

Private goods (excludable + rival): The seller can charge a price, collect revenue, and cover costs. When you buy a hotel room or an airline seat, no one else can use that room or seat simultaneously, and you can be turned away if you don't pay. Markets work well here.

Club goods (excludable + non-rival): A Netflix subscription does not become less valuable because millions of others also subscribe — your viewing does not reduce anyone else's. But Netflix *can* exclude non-subscribers behind a paywall. This excludability is precisely why the market can provide club goods efficiently: the firm charges an access fee to cover fixed costs, then lets subscribers consume freely (since MC per additional subscriber $\approx$ 0).

Common-pool resources (non-excludable + rival): A shared fishery is rival — each tonne of fish one boat catches is gone. But no boat can easily be excluded from the fishing grounds. The result is the **tragedy of the commons**: every individual has an incentive to fish as much as possible before others do, leading to collective over-exploitation and depletion far beyond the socially optimal harvest. Solutions include privatisation, quotas, or community management.

Pure public goods (non-excludable + non-rival): National defence protects every citizen simultaneously (non-rival), and once provided, no citizen can be excluded from its protec-

In plain English:
Two questions:
Can the seller exclude non-payers? (excludable/non-excludable). Does one person's use reduce others'? (rival/non-rival).

tion (non-excludable). Because no one can be charged for consuming it, the private market cannot recover costs — the free-rider problem prevents provision even when aggregate willingness to pay exceeds the cost.

Check Your Understanding

1. Classify each good, and justify your answer: (a) a private beach with a gate; (b) a public beach; (c) a satellite radio signal (SiriusXM subscription); (d) a city park on a quiet Tuesday; (e) a city park at a sold-out festival. For any “borderline” cases, explain what makes them borderline.
2. Why can markets efficiently provide club goods but not pure public goods, even though both are non-rival?
3. A tech company claims: “Our software is non-rival, so it should be free.” What is the economic error in this argument?

The key failure mode is non-excludability, which generates the free-rider problem. A game-theoretic lens shows exactly how this plays out.

“Non-rival means free.” Non-rivalry means the *marginal cost* of serving an additional consumer is zero, not that the good costs nothing to produce. National defence is non-rival, but costs hundreds of billions to provide. The non-rivalry property implies that *once the good exists*, everyone should have access at zero marginal charge — but the fixed cost of producing it must still be covered (usually through taxation).

11.2.3. The Free-Rider Problem

Free-Rider Problem: Because no one can be excluded from consuming a public good once it is provided, each individual has an incentive to wait for *someone else* to pay for it and then enjoy the benefits without contributing. In the extreme, if everyone reasons this way, the good is never provided even though everyone would benefit.

In words: The free-rider problem is the public-goods version of the prisoner’s dilemma: individual rationality produces collective irrationality.

A numerical illustration. Consider 3 neighbours who can jointly fund a neighbourhood security guard. The guard costs \$300/month. Each neighbour values the guard at \$120/month.

- **Is it socially optimal to hire the guard?** Total WTP = $3 \times \$120 = \$360 > \$300$. Yes — the guard generates a net social surplus of \$60.
- **Will anyone hire him?** Each neighbour reasons: “If the other two hire him, I benefit anyway at no cost. So I should free-ride.” If the other two split the cost equally (\$150 each), I save \$100 while still getting \$120 of protection.
- **Outcome:** Since every neighbour reasons identically, no one contributes. The guard is not hired. All three are worse off than if they had cooperated: the missed net surplus is \$60 (= \$360 – \$300).

To see this as a game, simplify to two neighbours A and B; cost \$200, each values the guard at \$120.

		Contribute (\$100)	Free-ride
A	Contribute	20, 20	−100, 120
	Free-ride	120, −100	0, 0

Payoff calculation: if both contribute \$100, each pays \$100 and gains \$120, net = +\$20. If A contributes alone, A pays \$100 to provide the guard; B free-rides for net \$120; A gets \$120 − \$200 = −\$80... but with two people sharing the cost equally at \$100 each, A alone bears the full \$200, getting 120 − 200 = −\$80. The matrix above uses the symmetric version where each contributes their \$100 share only if both do.

This is a **prisoner's dilemma**: “Free-ride” strictly dominates “Contribute” for each player. The unique Nash equilibrium is (Free-ride, Free-ride) = (0, 0), even though (Contribute, Contribute) = (20, 20) is Pareto superior. Individual rationality destroys the social surplus.

Check Your Understanding

1. Classify: (a) a congested highway, (b) a lighthouse, (c) a Netflix subscription, (d) groundwater from a shared aquifer.
2. Why does the free-rider problem not arise for private goods? (Hint: what role does excludability play?)
3. Give one real-world mechanism that governments use to overcome the free-rider problem, and explain how it changes the game.
4. True or false: “A Pigouvian subsidy can solve the free-rider problem for public goods.” Explain.

The free-rider problem prevents private markets from providing public goods. A game-theoretic analysis shows exactly why — and how partial contributions can transform the outcome.

11.3. Provision of a Public Good: Game-Theoretic Analysis

Two neighbours want a streetlight costing \$120. Amara values it at \$70, Ben at \$80. Neither values it enough to pay alone. What happens if they must each decide individually whether to pay?

11.3.1. Setup and Social Optimality

Let r_A and r_B be each person's reservation price, c the cost. The social optimality condition:

$$r_A + r_B > c.$$

In words: Sum everyone's willingness to pay. If the total exceeds cost, provision is socially desirable — even if no single person's WTP exceeds cost alone.

Vertical summation of demand for public goods (everyone benefits simultaneously) vs. horizontal summation for private goods (only

11.3.2. Case 3: Neither Can Afford It Alone, But Together They Can

This is the policy-relevant case: $r_A < c$, $r_B < c$, but $r_A + r_B > c$.

Binary Game: Buy-or-Not Decision

With $c = \$100$, $r_A = \$80$, $r_B = \$65$:

From Prisoner's Dilemma to Coordination Game

Binary game (each pays full c):

		Buy	Don't
A	Buy	-20, -35	-20, 65
	Don't	80, -35	0, 0

Don't dominates for both. **NE: (Don't, Don't)** = (0, 0). Good not provided. Pareto inferior to provision (surplus = \$45).

Partial-contribution game ($g_A = \$60$, $g_B = \$40$):

		Contrib.	Don't
A	Contrib.	20, 25	-60, 0
	Don't	0, -40	0, 0

Two NE: **(C,C)** = (20, 25) and **(D,D)** = (0, 0). Prisoner's dilemma $\Rightarrow$ **coordination game**.

Key insight: Partial contributions transform the game. Efficient provision becomes achievable as an NE. The challenge shifts from "will the good be provided?" to "can we coordinate on the good equilibrium?"

Allowing partial contributions changes the strategic structure from a prisoner's dilemma (dominant strategy not to contribute) to a coordination game (two Nash equilibria: the efficient one and the zero-provision one). A credible commitment mechanism — government mandates, social norms, crowdfunding with all-or-nothing thresholds — can coordinate expectations on the efficient equilibrium.

Kickstarter and the All-or-Nothing Rule. Crowdfunding platforms implement exactly the partial-contribution structure above. Backers pledge money but are only charged if the project reaches its full funding target. This design eliminates the dominant "don't contribute" strategy: a pledge costs nothing unless the target is met, so contributing is weakly dominant for any backer who values the project above zero. The all-or-nothing threshold transforms the public-good game into a coordination game. Real data show projects that reach $\approx 30\%$ of their goal in the first few days "tip" to full funding — consistent with the efficient coordination equilibrium becoming self-reinforcing once positive expectations form.

Check Your Understanding

1. Streetlight example: $c = \$120$, $r_A = \$70$, $r_B = \$80$, $g_A = g_B = \$60$. Verify the payoffs: (C,C) = (10, 20), (D,D) = (0, 0). Are both Nash equilibria? $\checkmark$
2. Why is (Don't, Don't) always a Nash equilibrium in the partial-contribution game?
3. What is the condition on g_i for (Contribute, Contribute) to also be a Nash equilibrium?

Public goods analysis reveals that free-riding leads to systematic under-provision. Now we turn to externalities, where production or consumption generates side-effects on third parties.

11.4. Externalities

A steel mill on a river produces steel that its customers value. It also produces effluent that kills fish downstream — fish the mill neither owns nor pays for harming. How does this change the efficiency of the market outcome?

Externality: A cost or benefit imposed on a third party not involved in the market transaction that generates it, and not reflected in market prices.

Negative externality: Imposes uncompensated costs $\Rightarrow SMC > PMC \Rightarrow$ overproduction.

Positive externality: Confers uncompensated benefits $\Rightarrow SMB > MB \Rightarrow$ underproduction.

In words: With a negative externality: the market produces too much because producers don't pay the full social cost. With a positive externality: the market produces too little because producers don't capture the full social benefit.

Negative externality examples: Factory air pollution (costs health damages not borne by the factory); traffic congestion (each driver slows everyone else); second-hand smoke.

Positive externality examples: Vaccination (immunised individuals reduce transmission probability for others); education (spillovers through innovation and civic participation); a well-maintained property (raises neighbours' property values).

Key notation:

$SMC =$

$PMC + MEC.$

Market equates

$PMC = MB$; *social optimum requires*

$SMC = MB.$

Check Your Understanding

1. For a negative externality: does the market equilibrium Q^E lie above or below the social optimum Q^* ? Why?
2. For a positive externality in production: does SMC lie above or below PMC ? What policy corrects the distortion?
3. True or false: "A production externality shifts the supply curve; a consumption externality shifts the demand curve."

11.5. Market Equilibrium with Externalities

The market equilibrium equates $PMC = MB$ and produces $Q^E = 45$ units. The socially efficient output is only $Q^* = 35$. How large is the welfare loss from the market's overproduction?

Consider a steel mill that dumps waste into a river. The mill's managers decide how much steel to produce by comparing the price they receive to their own production cost (PMC). They do not pay for the damage their effluent causes to downstream fishermen. This gap between what the mill pays and what society actually bears is the externality wedge: $SMC = PMC + MEC$, where MEC is the per-unit damage imposed on the fishermen. Because the mill ignores MEC , it equates $PMC = MB$ rather than $SMC = MB$, and produces *more*

than the social optimum. Every unit between Q^* and Q^E has a social cost (SMC) that exceeds its social benefit (MB) — that is the source of the deadweight loss.

11.5.1. Key Relationships

$$\text{SMC} = \text{PMC} + \text{MEC} \quad (\text{negative externality}) \quad (11.1)$$

$$\text{SMB} = \text{MB} + \text{MEB} \quad (\text{positive externality}) \quad (11.2)$$

Competitive equilibrium: $\text{PMC} = \text{MB}$ at Q^E (overproduction if negative externality).

Social optimum: $\text{SMC} = \text{MB}$ (or $\text{PMC} = \text{SMB}$ for positive externality) at Q^* .

Deadweight loss: Triangle between SMC and MB over the interval $[Q^*, Q^E]$.

Confusing MEC with SMC. The DWL triangle uses the distance between SMC and MB, not between MEC and anything else. $\text{DWL} = \frac{1}{2}(Q^E - Q^*) \times [\text{SMC}(Q^E) - \text{MB}(Q^E)]$. A common error is to compute the triangle as the area under the MEC curve alone.

11.5.2. Benchmark: Verified Numbers

$P = 100 - Q$, $\text{PMC} = 10 + Q$, $\text{MEC} = 20$ (constant):

	Competitive equilibrium Q^E	Social optimum Q^*
Condition	$\text{PMC} = \text{MB}: 100 - Q = 10 + Q$	$\text{SMC} = \text{MB}: 100 - Q = 30 + Q$
Quantity	$Q^E = 45$	$Q^* = 35$
Price	$P^E = \$55$	$P^* = \$65$
$\text{DWL} = \frac{1}{2}(45 - 35) \times [\text{SMC}(45) - \text{MB}(45)] = \frac{1}{2}(10)(20) = \$100 \checkmark$		

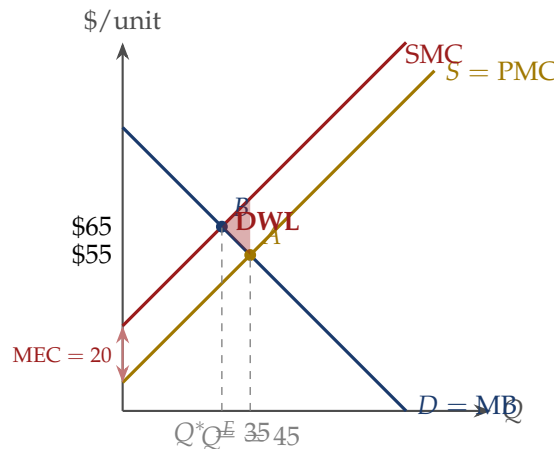


Figure 11.1. Negative externality: overproduction and DWL. *What to look for:* Market equilibrium A ($Q^E = 45$) lies right of the social optimum B ($Q^* = 35$). The DWL triangle (red) has base $= Q^E - Q^* = 10$ and height $= \text{SMC}(Q^E) - \text{MB}(Q^E) = 20$. Area $= \$100$. The Pigouvian tax shifts the supply curve up by $t^* = \$20$ to coincide with SMC, moving equilibrium to B.

Check Your Understanding

1. Using the benchmark ($P = 100 - Q$, $PMC = 10 + Q$, $MEC = 20$): verify that $Q^E = 45$, $Q^* = 35$, and $DWL = \$100$. Draw a labelled diagram showing the two equilibria and the DWL triangle.
2. If MEC doubled to \$40 per unit (but remained constant), what would the new social optimum be? How would the DWL change?
3. A production process generates $MEC = 3Q$. Without any calculation, explain why the social optimum will be further below the competitive equilibrium than in a case where MEC is constant. What does the slope of MEC imply for the Pigouvian tax?

Application: Business-Relevant Externalities.

1. **Tech platform network externalities (positive).** When a new user joins Uber, LinkedIn, or WhatsApp, every existing user benefits: there are more drivers, more potential connections, more people to message. This is a *positive externality* — the new user's decision to join confers unpriced benefits on all existing users. The market equilibrium under-supplies users relative to the social optimum because the joining user captures only her own private benefit, not the aggregate benefit she generates for others. This insight is the economic justification for why platforms subsidise early users: free trials, referral bonuses, and even paying users to join (as Uber did in new cities) are Pigouvian-style instruments that push adoption toward the social optimum. A platform charging nothing for access is not a pricing mistake — it is a rational response to positive network externalities.
2. **Corporate carbon emissions (negative).** A cement factory emitting CO₂ imposes costs on society through climate change — costs borne by people who have no connection to the factory's transactions. Without regulation, the factory equates $PMC = MB$ and over-produces relative to the social optimum. This is the textbook case for a Pigouvian carbon tax or cap-and-trade permit system. The "social cost of carbon" — the MEC of one tonne of CO₂ — is estimated by integrated assessment models and is the theoretical foundation for carbon pricing. By 2023, the EU carbon price exceeded €80/tonne, large enough to drive meaningful fuel-switching and investment in renewables in covered sectors.

We have identified the problem. Three policy instruments can solve it.

11.6. Solutions to the Externality Problem

A factory pollutes a river. Three economists propose three solutions: (1) assign property rights; (2) tax the factory; (3) cap and trade emissions. Are these equivalent? When does each work best?

11.6.1. Property Rights and the Coase Theorem

Coase Theorem (Coase, 1960): If property rights are well-defined, transaction costs are zero, and parties can bargain freely, then the market will reach an efficient allocation regardless of the initial assignment of property rights.

Efficient pollution level S^* where $MB_{\text{polluter}}(S^*) = MD_{\text{victim}}(S^*)$ is reached whether the right to clean air belongs to the victim or the polluter.

In words: Efficiency is achieved regardless of who gets the property right. Only the distribution of gains changes. The factory pollutes the same socially optimal amount either way.

Why the Coase theorem often fails in practice. The theorem is a powerful theoretical benchmark, but its three conditions are routinely violated:

*Coase benchmark:
zero transaction
costs + well-defined
rights $\Rightarrow$ negotiate
to efficiency.*

1. **Transaction costs.** When thousands or millions of people are affected — downstream communities, residents of a city, citizens of the world — negotiating a Coasian bargain is prohibitively expensive. The legal, logistical, and communication costs of assembling the affected parties, aggregating their preferences, and enforcing an agreement can easily exceed the surplus from the bargain itself.
2. **Hold-up and strategic behaviour.** Once one party makes a sunk investment (e.g., the factory builds a production facility), the other party gains bargaining leverage. The fishery may demand compensation far above its actual damages, knowing the factory cannot easily exit. This *hold-up problem* causes bargaining to break down or produce inefficient outcomes.
3. **Information asymmetry.** The polluter knows its true abatement cost; the victims do not. Victims know the extent of their damages; the polluter does not. Each side has incentives to misrepresent its private information to gain bargaining advantage, potentially preventing agreement even when mutual gains exist.
4. **Free-riding among victims.** If many downstream communities are harmed, each community may try to free-ride on the negotiating efforts of others, undermining the collective bargaining needed to extract compensation from the polluter.

When Coase Works — and When It Does Not. A factory and a fishery share a river. There are only two parties. Property rights are clear (say, the fishery has the right to clean water). Transaction costs are low — the factory and the fishery can negotiate directly. In this case, the Coasian prediction is reliable: the two parties will bargain to the efficient pollution level S^* , with the factory compensating the fishery for damages.

Now change the setting: the factory's emissions affect 500 downstream communities spread across three countries. The victims speak different languages, have different legal systems, and must first organise themselves before they can negotiate. Each community has an incentive to free-ride on others' efforts. Transaction costs are enormous. The Coase theorem's conditions fail catastrophically — exactly the situation that describes global carbon emissions. In these cases, unilateral government intervention (taxes, regulation, cap-and-trade) is the only practical path to the social optimum.

When Coasian bargaining is impractical, the government must directly alter incentives. The Pigouvian tax is the most analytically elegant instrument.

11.6.2. Pigouvian Taxes

Optimal Pigouvian Tax:

$$t^* = \text{MEC}(Q^*) \quad (11.3)$$

Set the per-unit tax equal to the marginal external cost *at the social optimum* Q^* , not at Q^E . The tax shifts the effective supply curve from PMC to $\text{PMC} + t^* = \text{SMC}$, moving equilibrium to the social optimum.

In words: The tax makes the polluter face the full social cost. At the correct tax level, the market equilibrium coincides with the social optimum.

Evaluating the Pigouvian tax at Q^E instead of Q^* . The correct tax is $t^* = \text{MEC}(Q^*)$, evaluated at the social optimum. When MEC is constant, this distinction is irrelevant. But when $\text{MEC} = bQ$, we must first solve for Q^* from $\text{SMC}(Q^*) = \text{MB}(Q^*)$, then evaluate MEC there: $t^* = bQ^*$.

Verified: $P = 200 - 3Q$, $\text{PMC} = 20 + Q$, $\text{MEC} = 2Q$. Social optimum: $Q^* = 30$. Tax: $t^* = 2(30) = \$60$ (not $\text{MEC}(Q^E) = 2(45) = \90).

A common error: evaluate MEC at the social optimum Q^* , not at the competitive equilibrium Q^E . The tax is designed to move the equilibrium *from* Q^E *to* Q^* , so the relevant marginal damage is that prevailing at the destination. When MEC is constant, this distinction is irrelevant. But when $\text{MEC} = bQ$, we must first solve $\text{SMC}(Q^*) = \text{MB}(Q^*)$ to find Q^* , then set $t^* = b \cdot Q^*$.

Numerical verification using the benchmark. $P = 100 - Q$, $\text{PMC} = 10 + Q$, $\text{MEC} = 20$ (constant). Social optimum: $100 - Q = 30 + Q \Rightarrow Q^* = 35$. Tax: $t^* = \text{MEC}(Q^*) = 20$. With tax, effective supply = $\text{PMC} + t^* = 10 + Q + 20 = 30 + Q = \text{SMC}$. New equilibrium: $100 - Q = 30 + Q \Rightarrow Q = 35 = Q^*$. The tax eliminates the DWL of \$100 entirely. ✓

Pigouvian subsidy for positive externalities: $s^* = \text{MEB}(Q^*)$. The subsidy shifts the effective demand curve upward by s^* , increasing equilibrium quantity to the social optimum.

Pigouvian taxes require the regulator to know MEC precisely — a demanding informational requirement. Cap-and-trade offers an alternative that achieves efficiency using the price mechanism even when the regulator knows only the target quantity, not the marginal damage function.

11.6.3. Cap-and-Trade Systems

Cap-and-Trade: The regulator sets a total pollution cap, issues permits equal to that cap, and allows firms to buy and sell permits. At equilibrium, all firms equate their marginal abatement cost to the permit price λ^* .

Efficient condition: $\text{MAC}_A(e_A^*) = \text{MAC}_B(e_B^*) = \lambda^*$ (equal marginal cost of abatement across all firms).

In words: Firms with low abatement costs reduce more and sell permits. Firms with high abatement costs reduce less and buy permits. The environmental target is hit at minimum total cost.

The core intuition. A regulator issues permits equal to the socially optimal output Q^* . Every firm now faces a market price for the right to emit: the permit price λ^* . Firms that can clean up cheaply (low MAC) will reduce more than required and *sell* their spare permits

— they profit from abatement. Firms with high abatement costs will *buy* permits rather than clean up. Trading continues until every firm's marginal cost of abatement equals the common permit price: $MAC_A = MAC_B = \lambda^*$. At this point, the total pollution target is met, and the total cost of achieving it is minimised. Equal regulation (same abatement for every firm) cannot achieve this because it forces high-cost firms to abate just as much as low-cost firms, squandering the cost advantage of the latter.

Verified Example: $MAC_A = 2e_A$, $MAC_B = 8e_B$, **total abatement** = 100.

Equal regulation ($e_A = e_B = 50$): Cost = $\frac{1}{2}(2)(2500) + \frac{1}{2}(8)(2500) = 2,500 + 10,000 = \$12,500$.

Cap-and-trade: equalise MACs: $2e_A = 8e_B \Rightarrow e_A = 4e_B$; with $e_A + e_B = 100$: $e_A^* = 80$, $e_B^* = 20$.

Cost = $\frac{1}{2}(2)(6400) + \frac{1}{2}(8)(400) = 6,400 + 1,600 = \$8,000$. Savings = \$4,500 (36%). Permit price $\lambda^* = 2(80) = 8(20) = \160 . ✓

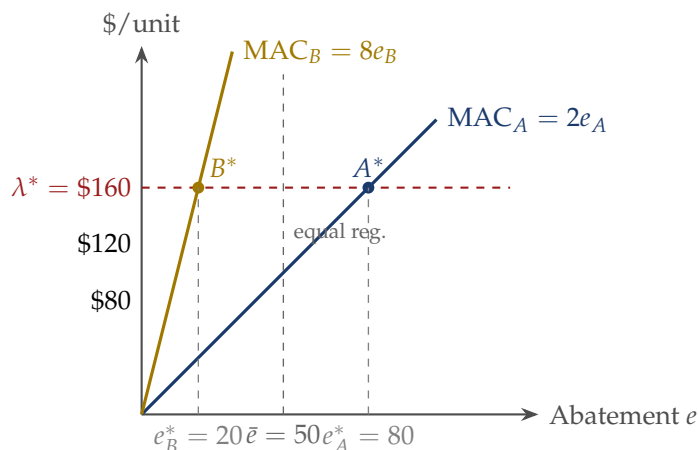


Figure 11.2. Cap-and-trade: MAC equalisation. *What to look for:* Firm A has a low MAC (flat slope); Firm B has a high MAC (steep slope). The equilibrium permit price $\lambda^* = \$160$ is shown as the horizontal dashed red line. Firm A abates 80 units (where MAC_A intersects λ^*); Firm B abates only 20 units (where MAC_B intersects λ^*). The grey dashed line at $\bar{e} = 50$ shows what equal regulation would require — Firm A would under-abate relative to its efficient level and Firm B would over-abate. Total cost under cap-and-trade (\$8,000) is \$4,500 less than under equal regulation (\$12,500).

Cap-and-trade achieves the environmental target at minimum cost. But choosing between a tax and a cap depends on more than economics — political economy matters enormously.

11.6.4. Comparing the Three Instruments

Policy Instrument Decision Guide

	Coase bargaining	Pigouvian tax	Cap-and-trade
Mechanism	Private negotiation	Per-unit tax on emissions	Tradeable permits
Info needed	Parties' own valuations	Regulator knows MEC	Regulator knows cap target
Achieves efficiency?	Yes (if conditions hold)	Yes (at correct t^*)	Yes
Revenue?	No (private transfer)	Yes (tax revenue)	Only if auctioned <i>Weitz-</i>
Price certainty	Uncertain	Yes (price fixed at t^*)	No (price fluctuates)
Quantity certainty	Uncertain	No (quantity responds)	Yes (cap is binding)
Works when	Few parties, low TX costs	MEC well-measured	Many firms, diverse costs
Breaks down when	Many parties, high TX costs	Uncertainty about MEC	Permit price volatility

man (1974): use a tax (price instrument) when marginal abatement costs are steep; use a cap (quantity instrument) when marginal damages are steep (e.g., cliff-edge damage thresholds).

Political economy of instrument choice. Carbon taxes are theoretically simpler — a single number (t^*) summarises the policy — but politically costly: voters see the tax line on their energy bill and resist it. Governments that have tried to raise carbon taxes have faced fuel protests, electoral backlash, and policy reversals. Cap-and-trade is politically easier for two reasons: first, the “cost” of pollution is the opportunity cost of the permit (foregone revenue from selling it), which is less visible to consumers than an explicit tax; second, permits can be distributed free to existing firms (*grandfathering*), buying political support from the industries that would otherwise lobby hardest against the policy. The EU ETS used grandfathering extensively in its early phases (2005–2012), which is why early permit prices were too low to drive significant abatement. As auctioning has replaced grandfathering post-2012, both the price signal and the government revenue from the ETS have increased substantially. Real-world carbon pricing systems often combine elements of both instruments: an explicit price floor (minimum carbon price) combined with a tradeable permit system, providing both the price certainty of a tax and the quantity certainty of a cap.

Application: Carbon Pricing in Practice — Tax vs. Cap-and-Trade.

EU Emissions Trading System (cap-and-trade): Launched 2005, covering $\approx 40\%$ of EU emissions from power generation and large industry. Permits were initially grandfathered (no revenue); auctioning has increased over time. Early permit prices were too low ($< \text{€}10/\text{tonne}$) to drive significant abatement; prices rose sharply post-2018, reaching $> \text{€}80/\text{tonne}$ by 2023, significantly reducing covered-sector emissions.

British Columbia carbon tax (price instrument): Introduced in 2008 at C\$10/tonne CO_2 , rising gradually to C\$65/tonne by 2023 (scheduled to reach C\$170/tonne by 2030). Revenue

is recycled as dividends and income-tax cuts. Studies show BC's tax reduced gasoline consumption by $\approx 15\%$ relative to the rest of Canada. It is widely regarded as the world's most successful carbon tax to date.

Comparison: BC's tax gives firms price certainty (they know exactly what emissions cost, facilitating long-run investment). The EU ETS gives quantity certainty (total emissions are capped). BC's approach generates clear revenue; the EU ETS generates revenue only when permits are auctioned. Neither dominates universally — the choice depends on the steepness of the marginal damage curve (Weitzman).

Check Your Understanding

1. Pigouvian tax: $P = 100 - Q$, $PMC = 10 + Q$, $MEC = 20$. Find t^* and verify it moves equilibrium to $Q^* = 35$.
2. Under a Pigouvian tax, does the firm stop polluting entirely? Explain why or why not.
3. Cap-and-trade saves \$4,500 vs. equal regulation above. Why? What economic principle drives this cost saving?
4. In the cap-and-trade example with $MAC_A = 2e_A$, $MAC_B = 8e_B$, if the regulator issues too many permits (say, total abatement falls to only 50 instead of 100), what happens to the equilibrium permit price λ^* and the level of pollution?
5. Under what conditions does the Coase theorem produce the social optimum even when there is a negative externality? Give a real-world example where each condition holds, and one where it fails.

Before the practice questions, worked examples consolidate the three benchmarks.

11.7. Solved Examples

How do the three policy instruments — property rights bargaining, Pigouvian tax, and cap-and-trade — each restore the social optimum in concrete numerical settings? Work through these examples before attempting the practice questions.

Example 11.1: Streetlight (Public Good).

$c = \$120$, $r_A = \$70$, $r_B = \$80$, $g_A = g_B = \$60$.

- (a) Social desirable? $70 + 80 = \$150 > \120 . Yes.
- (b) Payoff matrix: $(C,C) = (10, 20)$; $(C,D) = (-60, 0)$; $(D,C) = (0, -60)$; $(D,D) = (0, 0)$.
- (c) Nash equilibria: both (Contribute, Contribute) and (Don't, Don't). Coordination game.
- (d) Without a mechanism to coordinate on (C,C), zero provision may result despite social desirability.

Example 11.2: Pigouvian Tax.

$P = 100 - Q$, $PMC = 10 + Q$, $MEC = 20$ (constant).

- (a) Competitive equilibrium: $PMC = MB \Rightarrow Q^E = 45$, $P^E = \$55$.
- (b) Social optimum: $SMC = 30 + Q = MB \Rightarrow Q^* = 35$, $P^* = \$65$.
- (c) $t^* = MEC(Q^*) = \$20$. Effective supply: $30 + Q$. New equilibrium: $Q^* = 35$. ✓
- (d) $DWL = (1/2)(10)(20) = \$100$ eliminated by the tax.

Example 11.3: Cap-and-Trade.

Three firms each pollute 600 units; regulator targets 900 total. MACs: $MAC_1 = 3e_1$, $MAC_2 = 6e_2$, $MAC_3 = 9e_3$.

Equal regulation (each abates 300): Cost = $\$135,000 + \$270,000 + \$405,000 = \$810,000$.

Cap-and-trade (equalise MACs at λ): $3e_1 = 6e_2 = 9e_3$ and $e_1 + e_2 + e_3 = 900$. Solution: $e_1 \approx 491$, $e_2 \approx 245$, $e_3 \approx 164$. Total cost $\approx \$663,000$. Savings $\approx \$147,000$ (18%).

Where Does This Take Us?

- **Network externalities (Q23):** Positive demand-side externalities where each additional user raises value for all existing users. Social media, telecoms, and payment systems all exhibit this property — markets systematically under-provide these goods relative to the social optimum, justifying subsidised early adoption.
- **The double-dividend hypothesis:** Pigouvian tax revenue can fund reductions in other distortionary taxes (income taxes, payroll taxes), yielding a second efficiency gain beyond the externality correction. This is a key argument for carbon taxes with dividend recycling.
- **Weitzman (1974) instrument choice:** With steep marginal damage (climate “tipping points”), quantity instruments (caps) are preferred. With steep MAC curves, price instruments (taxes) are preferred. This insight is central to the carbon policy debate.
- **Climate economics:** The analysis in this chapter is the intellectual foundation for all carbon pricing policy. Integrated Assessment Models estimate the MEC of CO₂ (the “social cost of carbon”) — the number that determines the optimal Pigouvian tax.

Five Common Exam Mistakes — and How to Avoid Them

1. **Using $MEC(Q^E)$ instead of $MEC(Q^*)$ for the Pigouvian tax.** The optimal tax is $t^* = MEC(Q^*)$, evaluated at the social optimum. For constant MEC, this distinction is irrelevant. But for $MEC = bQ$, you must first find Q^* from $SMC(Q^*) = MB(Q^*)$, then calculate $t^* = b \cdot Q^*$. Using Q^E gives the wrong (too large) tax.
2. **Computing DWL as the area under the MEC curve.** The DWL triangle lies between the SMC and MB curves over $[Q^*, Q^E]$. Its base is $Q^E - Q^*$ and its height is $SMC(Q^E) - MB(Q^E)$. The MEC curve is not directly relevant to the DWL calculation.
3. **Claiming the Coase theorem works regardless of transaction costs.** The theorem requires zero transaction costs, well-defined property rights, and full information. With millions

of affected parties (global carbon emissions), bargaining is practically impossible. Always state the conditions when applying the theorem.

4. **Equating “non-rival” with “free.”** Non-rivalry means the marginal cost of serving an additional consumer is zero — not that the good costs nothing to produce. National defence is non-rival but enormously costly. The policy implication is that once the good *exists*, everyone should access it at zero marginal charge; but the fixed cost of provision must still be covered.
5. **Claiming cap-and-trade always generates government revenue.** Revenue is generated only if permits are *auctioned*. Grandfathered permits (allocated free to existing firms based on historical emissions) generate no government revenue and may create windfall profits for firms. The EU ETS initially used grandfathering extensively, which reduced the system’s fiscal benefit. When a question asks about government revenue from cap-and-trade, check how permits are distributed.
6. **Applying MEC at the competitive output Q^E instead of the social optimum Q^* when computing the Pigouvian tax t^* .** The Pigouvian tax is not designed to penalise production at the status quo — it is designed to *move* production from Q^E to Q^* . The incentive needed is the external damage at the destination, not the origin. For constant MEC this error is harmless, but for $MEC = bQ$, using Q^E gives a tax that is systematically too large.
7. **Confusing the Coase theorem with government regulation.** The Coase theorem is a result about *private bargaining*: in the absence of transaction costs and with well-defined property rights, affected parties negotiate to the efficient outcome *without government help*. A Pigouvian tax or cap-and-trade is a *government* intervention. The two approaches are substitutes: when Coasian bargaining is feasible, the government need not intervene; when it is not (many parties, high costs), government instruments are needed. Never describe a Pigouvian tax as an application of the Coase theorem.

Chapter at a Glance — Week 11: Public Goods and Externalities

Your one-page revision guide.

Key Equations

$r_A + r_B > c$	Social optimality: public good
$SMC = PMC + MEC$	Neg. externality
$SMB = MB + MEB$	Pos. externality
$t^* = MEC(Q^*)$	Pigouvian tax
$MAC_A = MAC_B = \lambda^*$	C&T equilibrium

Policy Instruments

	Revenue?	Qty certainty?
Coase bargaining	No	No
Pigouvian tax	Yes	No
C&T (auctioned)	Yes	Yes
C&T (grandfathered)	No	Yes

Benchmark: $P = 100 - Q$, $PMC = 10 + Q$, $MEC = 20$
 $Q^E = 45$, $P^E = \$55$; $Q^* = 35$, $P^* = \$65$; $t^* = \$20$;
 DWL = \$100 ✓

Public Good Game Summary

Game	Structure	NE
Binary (pay full c)	Prisoner's dilemma	(D,D) only
Partial contribution	Coordination game	(C,C) and (D,D)

Key: partial contributions enable efficient provision as an NE.

Four-Quadrant Goods

	Rival	Non-rival
Excludable	Private	Club
Non-excludable	Common-pool	Public

LO Checklist

- LO1: Taxonomy + free-rider problem ✓
- LO2: PD → coordination game (partial contributions) ✓
- LO3: $DWL = (1/2)(Q^E - Q^*)(SMC - MB)$ ✓
- LO4: Coase theorem + conditions ✓
- LO5: $t^* = MEC(Q^*)$ and DWL elimination ✓
- LO6: C&T efficiency: equalise MACs ✓

C&T Verified Example $MAC_A = 2e_A$,
 $MAC_B = 8e_B$, total = 100:
 $e_A^* = 80$, $e_B^* = 20$, $\lambda^* = \$160$
 Cost: \$8,000 vs. \$12,500 (equal reg.)
 Savings: \$4,500 (36%) ✓

The thread: Non-excludability $\Rightarrow$ free-rider problem $\Rightarrow$ binary game is a prisoner's dilemma (no provision) $\rightarrow$ partial contributions transform to coordination game (two NE) $\rightarrow$ negative externality: $PMC < SMC \Rightarrow Q^E > Q^*$, DWL $\rightarrow$ three remedies: Coase (if low TX costs), Pigou tax ($t^* = MEC(Q^*)$ internalises), cap-and-trade (efficient abatement by equalising MACs) $\rightarrow$ choice between tax and cap depends on steepness of MAC vs. marginal damage curves (Weitzman).

11.8. Key Terms

Non-excludability

Once provided, no individual can be prevented from consuming the good; root cause of the free-rider problem.

Non-rivalry

One person's consumption does not reduce others' availability; MC of additional consumer = 0.

Pure Public Good

Non-excludable and non-rival simultaneously (e.g., national defence, broadcast radio).

Club Good

Non-rival but excludable (cable TV, toll roads, gym membership).

Common-Pool Resource

Rival but non-excludable (fisheries, groundwater); subject to tragedy of the commons.

Free-Rider Problem

Incentive to consume without contributing because exclusion is impossible; leads to private under-provision.

Vertical Summation

Aggregation of WTP for a public good: all consumers' WTP are summed (all benefit simultaneously). Contrast with horizontal summation for private goods.

Externality

Cost or benefit falling on a third party outside the market transaction, not reflected in prices. *Used throughout the course.*

Negative Externality

External cost $\Rightarrow$ $SMC > PMC \Rightarrow$ overproduction.

Positive Externality

External benefit $\Rightarrow$ $SMB > MB \Rightarrow$ under-

production.

PMC, MEC, SMC

Private marginal cost; marginal external cost; social marginal cost = $PMC + MEC$.

Coase Theorem

With well-defined rights, zero transaction costs, and free bargaining, efficient outcome is achieved regardless of initial rights assignment. *Used throughout the course.*

Pigouvian Tax

Per-unit tax $t^* = MEC(Q^*)$ at the social optimum; internalises the externality by aligning private and social costs.

Pigouvian Subsidy

Per-unit subsidy $s^* = MEB(Q^*)$ for positive externalities; corrects underproduction.

Cap-and-Trade

Regulator sets a cap; permits traded so abatement goes to lowest-cost firms; achieves target at minimum cost when $MAC_A = MAC_B = \lambda^*$.

Marginal Abatement Cost (MAC)

Cost of reducing pollution by one more unit.

Double-Dividend Hypothesis

Pigouvian tax revenue used to cut other distortionary taxes yields two efficiency gains: externality correction + tax reduction.

Network Externality

Positive demand-side externality: each additional user raises value for all existing users (social media, telecoms, payment systems).

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter.

1. A Deeper Look at Public Goods (MRU — *Marginal Revolution University*) ~ 9 min

<https://www.youtube.com/watch?v=hA2z-X31IvI>

Covers non-excludability, non-rivalry, the four-quadrant taxonomy, and the free-rider problem with clear examples. Reinforces Sections 11.2–11.3 and recommended before Level 1–2 public goods questions.

2. Externalities and Pigouvian Taxes (Jacob Clifford, *ACDC Economics*) ~ 11 min

<https://www.youtube.com/watch?v=ha-ssol6S0Q>

Derives the negative externality diagram (PMC, MEC, SMC), identifies the competitive equilibrium vs. social optimum, and shows how a Pigouvian tax corrects the distortion. Reinforces Sections 11.5–11.6.

3. The Coase Theorem and Negative Externalities (Edspira) ~ 8 min

<https://www.youtube.com/watch?v=h2022KtjSck>

Explains the Coase theorem with numerical examples, demonstrates that the efficient outcome is the same regardless of rights assignment, and discusses the theorem's limitations (transaction costs, free-riding).

4. Cap-and-Trade vs. Carbon Tax (McCombs School of Business) ~ 14 min

<https://www.youtube.com/watch?v=eTjHzulYkeU>

Compares cap-and-trade and Pigouvian taxes on multiple dimensions (price certainty, quantity certainty, revenue, distributional effects), and uses real-world policy examples as case studies. Reinforces Section 11.6 and the decision guide.

11.9. Practice Questions

How to use these questions. Difficulty: ★ easy ★★ intermediate ★★★ challenging.
Attempt every question independently before consulting solutions.

Level 1: Conceptual and Short-Answer

- ★ **[Taxonomy of Goods]** Define non-excludability and non-rivalry. For each good below, state which properties are present and classify as private, club, common-pool, or public good: (a) congested highway; (b) lighthouse; (c) Netflix subscription; (d) groundwater from a shared aquifer.
- ★ **[Free-Rider Problem]** Explain the free-rider problem. Why does it prevent the private market from providing the socially optimal quantity of a public good even when everyone would benefit?
- ★ **[Binary Public-Good Game]** $r_X = \$50$, $r_Y = \$70$, $c = \$90$.
 - (a) Is provision socially desirable?
 - (b) Derive the payoff matrix (binary buy-or-not) and find the Nash equilibrium.
 - (c) Does the NE achieve the social optimum? Why not?
- ★ **[Partial Contributions]** Continue Q3: $g_X = \$40$, $g_Y = \$50$ (good provided only if both contribute). Derive the payoff matrix and find all Nash equilibria.
- ★ **[Externality Concepts]** Distinguish negative externalities in production from those in consumption. Give one real-world example of each. In which case does the supply curve equal the PMC?
- ★ **[Coase Theorem]**
 - (a) State the Coase theorem precisely, including all required conditions.
 - (b) A steel mill discharges into a river, harming a downstream fishery. Rights assigned to the fishery: describe the Coasian bargain.
 - (c) Rights assigned to the mill: does the efficient outcome change? Does the distribution change?
 - (d) Give two settings where the Coase theorem works well, and two where it breaks down.
- ★ **[Benchmark — No Externality]** $P = 80 - 2Q$, $PMC = 4Q$. Find the competitive equilibrium and verify $DWL = 0$.
- ★ **[Explain to a Friend — Level 1]** *No formulas.* Your friend says: “Cap-and-trade sounds like a polluters’ free pass. Firms just pay to pollute. Why would an economist prefer this to just telling firms how much they can emit?” Write 3–4 plain-English sentences explaining the cost-efficiency argument for cap-and-trade.

★ **[Coase Theorem: Conditions and Failures]** The Coase theorem is often described as “the most important result in environmental economics.”

- (a) List the three conditions required for the Coase theorem to hold.
- (b) A factory and a single neighbouring farm share a drainage ditch. The factory’s effluent reduces the farm’s crop yield. Explain why the Coase theorem is likely to work well here.
- (c) Now suppose the same factory operates in a densely populated region and its emissions affect 2 million residents across four states. Identify which condition(s) of the Coase theorem fail, and explain *why* each fails.
- (d) What government policy instrument would you recommend in the second scenario? Briefly justify your choice.

Level 2: Intermediate Problems

★★ **[Pigouvian Tax — Negative Externality]** Continue Q7 market with $MEC = 12$ (constant).

- (a) Find the social optimum (Q^*, P^*) .
- (b) Compute the DWL at the competitive equilibrium.
- (c) Calculate t^* . Verify it achieves Q^* .

★★ **[Positive Externality — Pigouvian Subsidy]** $P = 120 - Q$, $PMC = 20 + Q$, $MEB = 15$ (constant).

- (a) Private market equilibrium.
- (b) Social optimum.
- (c) Pigouvian subsidy.
- (d) By how much does the market under-produce?

★★ **[Coase Theorem — Numerical]** Factory: $MB_F = 60 - 5S$. Laundry: marginal damage $MD_L = 3S$ (S = smoke units).

- (a) Factory’s private optimum.
- (b) Socially efficient smoke level.
- (c) Laundry owns clean air: describe the Coasian bargain.
- (d) Factory owns air: outcome and distribution.

★★ **[Cap-and-Trade — Efficient Allocation]** $MAC_A = 2e_A$, $MAC_B = 8e_B$; total abatement required = 100.

- (a) Equal regulation cost.

- (b) Efficient split (e_A^*, e_B^*) under cap-and-trade.
- (c) Cap-and-trade total cost.
- (d) Equilibrium permit price λ^* .

★★ **[Data-Based: Welfare Under Externalities]** Complete the table for a market with $P = 120 - Q$, $PMC = 20 + Q$, $MEC = 2Q$ (variable).

Scenario	Q	P	PMC	SMC	DWL
Competitive equilibrium					n/a
Social optimum					0
Tax $t = \$40$ applied					

- (a) Fill in Q , P , PMC , SMC for each scenario.
 - (b) Compute DWL at the competitive equilibrium and the residual DWL when $t = \$40$.
 - (c) What is the optimal Pigouvian tax t^* ? Why is $t = \$40$ too low?
 - (d) Draw a labelled supply/demand diagram showing all three scenarios.
 - (e) At the competitive equilibrium, how much of the DWL is a transfer and how much is pure waste? (Hint: decompose the area between SMC and MB .)
- ★★ **[Three-Person Public Good]** $v_1 = \$200$, $v_2 = \$150$, $v_3 = \$100$; cost = \$400. Contributions: $g_1 = \$210$, $g_2 = \$140$, $g_3 = \$50$.
- (a) Socially desirable to provide?
 - (b) Will each consumer voluntarily contribute?
 - (c) With g_3 fixed, what is the Nash equilibrium in the sub-game between C_1 and C_2 ?
- ★★ **[Cap-and-Trade — Two Firms (chapter data)]** Firms A and B each currently pollute 1,000 gallons. Each receives 500 permits. $MAC_A = \$10/\text{gallon}$, $MAC_B = \$2/\text{gallon}$.
- (a) At what permit price is B indifferent?
 - (b) At what price is A indifferent?
 - (c) Equilibrium permit price range and total clean-up cost.
- ★★ **[Gasoline Tax]** $P = 5 - 0.01Q$, $PMC = 1 + 0.005Q$ (dollars/litre, Q in millions). $MEC = \$0.50/\text{litre}$.
- (a) Competitive equilibrium.
 - (b) Social optimum.
 - (c) Pigouvian tax.

- (d) DWL eliminated by the tax.
- (e) Tax set too low at \$0.25: new equilibrium and residual DWL.

★★ [Explain to a Friend — Level 2] Your friend says: “I don’t understand why the optimal Pigouvian tax is $MEC(Q^*)$ and not $MEC(Q^E)$. Isn’t the problem happening at the market equilibrium?” Write a 4–5 sentence explanation of why the tax is evaluated at the social optimum.

Level 3: Advanced Problems

★★★ [Variable MEC — Full Analysis] $P = 200 - 3Q$, $PMC = 20 + Q$, $MEC = 2Q$.

- (a) Competitive equilibrium.
- (b) Social optimum.
- (c) DWL at competitive equilibrium.
- (d) Optimal Pigouvian tax.
- (e) Verify the tax achieves Q^* .

★★★ [General Partial Contributions] r_A, r_B, c general ($r_A < c, r_B < c, r_A + r_B > c$); $g_A + g_B = c$.

- (a) Payoff matrix for the partial-contribution game.
- (b) Condition for (Contribute, Contribute) to be a Nash equilibrium.
- (c) Show (Don’t, Don’t) is always a Nash equilibrium.
- (d) Can the efficient outcome ever be the *unique* NE?

★★★ [General Cap-and-Trade] n firms each emit $\bar{e}$; total permits $N < n\bar{e}$; $MAC_i = c_i a_i$.

- (a) Show abatement is inversely proportional to c_i at equilibrium.
- (b) Derive the equilibrium permit price λ^* .
- (c) With $n = 3$, $\bar{e} = 1,000$, $N = 1,500$, $c_1 = 2$, $c_2 = 4$, $c_3 = 6$: find each firm’s abatement and the permit price.
- (d) Compare total cost to equal abatement (500 each).

★★★ [Pigouvian Tax vs. Cap-and-Trade — Policy Analysis] “A Pigouvian tax is always superior to cap-and-trade because it generates government revenue.” Critically evaluate this claim, addressing: (a) the double-dividend hypothesis; (b) price vs. quantity certainty; (c) Weitzman’s (1974) instrument-choice result; (d) two advantages of cap-and-trade.

★★★ [Beekeepers and Positive Externalities] Three beekeepers: $MB_i = 80 - 4h_i$ (private), $MEB = \$10/\text{hive}$, $MC = \$50/\text{hive}$.

- (a) Private equilibrium hives per beekeeper.
- (b) Social optimum hives per beekeeper.
- (c) Pigouvian subsidy.
- (d) How might the Coase theorem resolve this? Practical barriers?
- (e) Total hives across all three beekeepers under each scenario.

★★★ **[Network Externalities]** Demand: $P = 120 - Q + 0.5N$ (N = installed base, Q = users). Initially $N = 0$; $PMC = 0$.

- (a) Competitive equilibrium with $N = 0$.
- (b) Social marginal benefit: $SMB = MB + MEB$ where $MEB \approx 0.5Q$.
- (c) Social optimum.
- (d) Gap between market equilibrium and social optimum. What policy could close it?
- (e) Explain why platforms like Facebook were initially free, using your results.

★★★ **[Explain to a Friend — Level 3]** Your friend, a policy student, says: “Climate change is the biggest externality in history. The obvious solution is a global Pigouvian carbon tax. Why hasn’t it happened?” Write a 5–6 sentence response addressing: (a) why the Coase theorem fails at the global scale; (b) why a Pigouvian tax is theoretically correct; (c) two practical obstacles to implementing a global carbon tax; (d) why cap-and-trade has been more politically feasible; and (e) what the Weitzman insight implies for instrument choice given climate “tipping points.”

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.